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Theme A FinSwarm

Architectural Topology & Graph State Management The boardroom engine utilizes a stateful graph topology constructed via LangGraph, replacing linear script execution with an iterative, cyclic decision-making engine. The core structure relies on a centralized BoardroomState typed dictionary that acts as a shared ledger. As execution flows from node to node, every department appends its analytical findings, risk parameters, and financial constraints into this shared state, ensuring complete transparency and auditability across the entire simulation lifecycle.

Multi-Agent Economic Reasoning & Friction Simulation

Unlike monolithic prompting where a single model attempts to balance conflicting corporate goals, this architecture isolates specialized personas to simulate real organizational friction:

- **Business Research Agent:** Parses arbitrary scenario descriptions to extract structural opportunities, market segment sizes (e.g., retail shops, service SMEs, small manufacturers), and demand metrics without chasing short-term revenue.
- **Finance and Treasury Agent:** Enforces strict unit economics, modeling capital allocation limits, cost of funds (e.g., 10% to 13%), servicing expenses, setup budgets, and mandatory liquidity reserves (such as the INR 3 crore floor).
- **Credit Risk Agent:** Calculates expected credit losses, evaluates segment-specific default thresholds (keeping portfolios strictly at or below 5% or 5.5%), monitors fraud vectors, and sets rigid verification rules.
- **Compliance and Customer Protection Agent:** Acts as an unyielding moral safeguard, enforcing transparent pricing, protecting against predatory terms, and maintaining interest rate caps (such as the 19% annual ceiling).
- **Marketing and Sales Agent:** Optimizes acquisition channels (partner accountants, digital ads, trade associations, referrals) within strict marketing budget caps.

Business Brief → Research → Finance → Credit Risk → Compliance → Marketing → Increment Cycle ↻ → CEO Decision

Iterative Debate & Review Protocol To prevent superficial consensus, the graph implements a conditional routing loop controlled by a review cycle counter (capped at three iterations). During these cycles, downstream agents cross-examine upstream constraints—for example, forcing Credit Risk and Finance to challenge aggressive marketing expansion plans or capital deployments that breach risk boundaries.

CEO Synthesis & Decision Output Once the review cycles conclude, control routes to the CEO Agent. The CEO synthesizes the accumulated debate trace, balances sustainable growth against credit loss exposure, and outputs a rigorous executive memorandum. This final document explicitly details the customer segment mix, product terms, approval policies, budget allocations, risk limits, go-to-market strategies, implementation sequences, and a dedicated section containing at least three quantifiable business KPIs.

Fault Tolerance & API Resilience

To ensure reliability during live execution or high-throughput evaluations, the system embeds a multi-key fail-safe wrapper (`invoke_with_key_switching`). If a primary Google Gemini API key encounters a rate limit (429) or runtime exception, the wrapper intercepts the error, pauses briefly, hot-swaps to a secondary API key index, and re-invokes the model seamlessly without terminating the workflow.